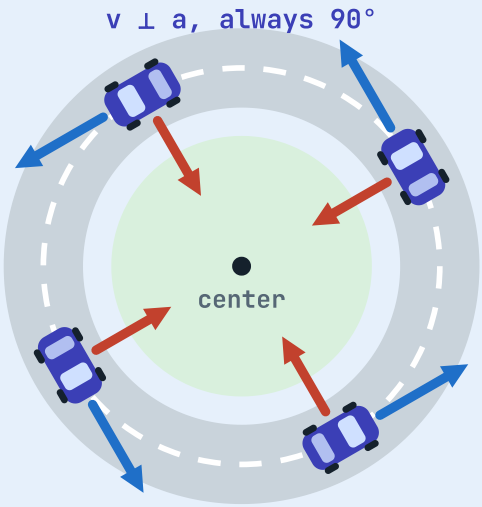
Circular Motion: The Big Idea

01

Going around a circle means you are **always turning**, so you are **always accelerating**, even at a steady speed.

Same speed. Different direction. → Accelerating!



velocity v
tangent: the way it's going

acceleration a
always toward the center

The speedometer never changes. The direction does.



Velocity → tangent

Points the way you're going right now: along the edge of the circle.



Acceleration → center

"Centripetal" = center-seeking.

$a = v^2/r$

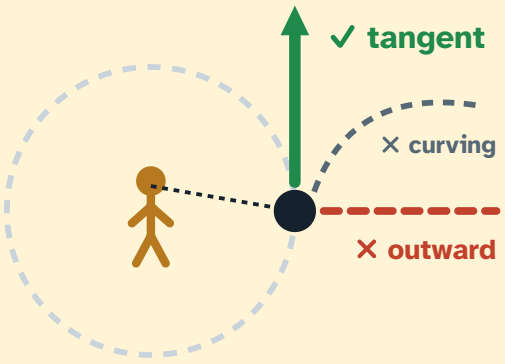


Net force → center

Some real force must pull inward.

$\Sigma F = mv^2/r$

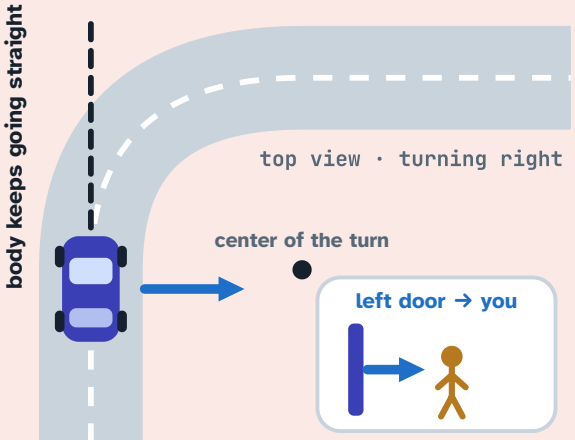
Let go! Which way does it fly?



top view · spinning counter-clockwise

No string = no force = **straight line** (Newton's 1st law), along the **tangent**. Never straight outward.

"Centrifugal force" is a myth



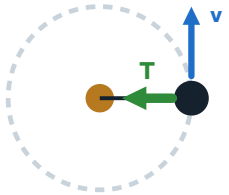
The car turns, but your body wants to keep going **straight**. The door runs into you and pushes you **inward**. Nothing pushes you out.

Who's Pulling Toward the Center?

02

"Centripetal force" is a **job**, not a new force. In every problem, find the **real** force (or forces) doing that job.

Ball on a string

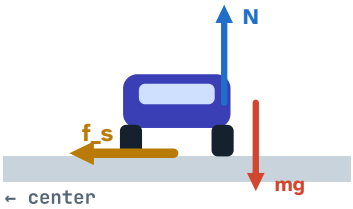


Tension

$T = mv^2/r$

Top view: the string pulls the ball toward your hand.

Car on a flat curve

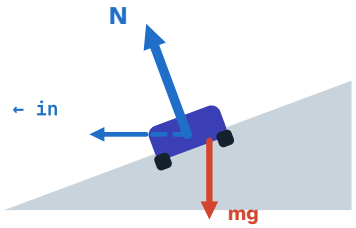


Static friction

$f_s = mv^2/r$

Rear view. On ice there's no friction, so the car slides straight.

Banked curve

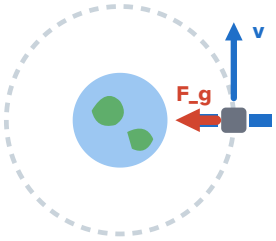


Part of the normal force

$N \sin\theta = mv^2/r$

The tilted road leans N toward the center: no friction needed at the design speed.

Moon or satellite

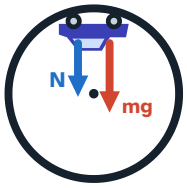


Gravity

$GMm/r^2 = mv^2/r$

Always falling toward Earth, moving sideways fast enough to keep missing.

Top of a coaster loop



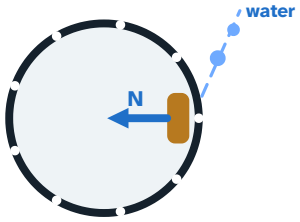
↓ center is below the car

Gravity + normal (both ↓)

$N + mg = mv^2/r$

Upside down, both forces point down to the center. Fast = the track pushes harder.

Washing-machine spin



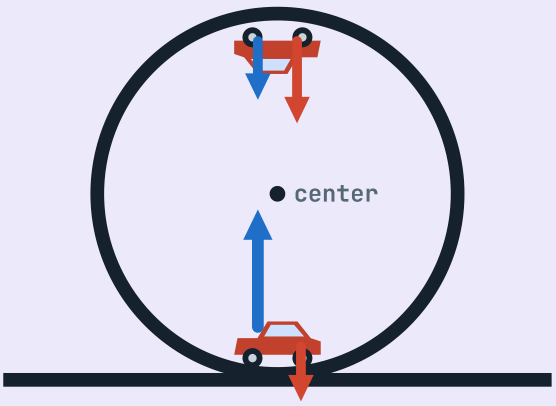
Normal force from the wall

$N = mv^2/r$

The wall pushes clothes inward. Water has nothing pushing it, so it escapes through the holes along the tangent.

The loop: heavy at the bottom, light at the top

N (blue) · mg (red)



TOP: both forces point down

$N + mg = mv^2/r$

feel light

BOTTOM: N must beat gravity

$N - mg = mv^2/r$

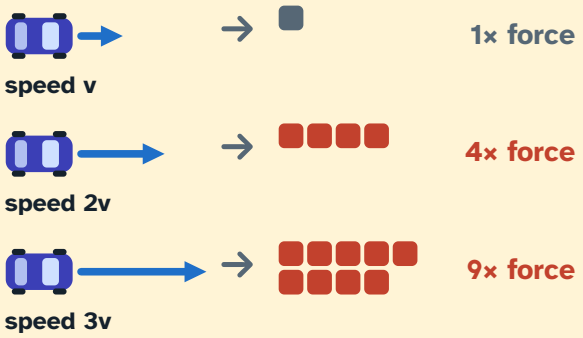
feel HEAVY

Numbers, Patterns & Test Traps

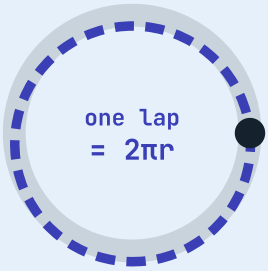
03

Squares, periods, hills and orbits, plus the recipe that earns FRQ points.

Speed is SQUARED



One lap, one period





takes T seconds

$v = 2\pi r / T$

$f = 1 / T$

Hill = lighter · Dip = heavier



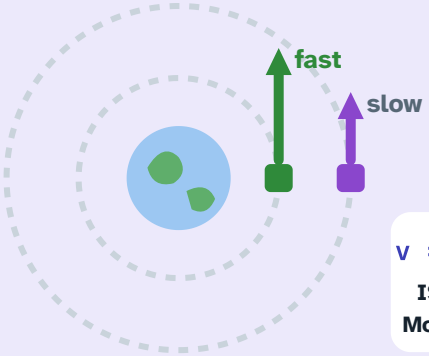
N < mg

lighter · airtime at $\sqrt{(gr)}$

N > mg

heavier

Closer orbit = faster



$v = \sqrt{(GM/r)}$

ISS: 90 min

Moon: 27 days

The 4-step recipe (works every time)

1. **Draw** — **REAL forces** — no F_c arrow
2. **Find** — **the center** — + = toward it
3. **ΣF** — **= mv²/r** — in +, out -
4. **Solve,** — **then check** — symbols first

Trap → Truth

✕ "Centrifugal force pushes me out"	→	✓ Inertia: you keep going straight
✕ Constant speed → a = 0	→	✓ Direction changes → a = v^2/r inward
✕ Let go → flies outward	→	✓ Flies along the tangent
✕ String tension points up at the top	→	✓ A string always pulls toward the center
✕ 2× speed → 2× force	→	✓ 2× speed → 4× force (v^2)